Model Dependence of PDG Comparison Values:

How Empirical Values Arise, Where Theory and Irrational Factors Enter, and What This Means for FFGFT Residuals

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Contents

Status markers

[K] *Confirmed* — derived from ξ, numerically verified.

[B] *Proved* — algebraically established.

[S] *Speculative* — open.

[Q] *Source* — corpus-external primary source, verified here.

Abstract

FFGFT predictions are compared against PDG and CODATA values. This document examines how those comparison values actually arise: which raw observable underlies each, which theory (QED, Standard Model) enters the extraction, and where irrational numbers already appear in the extraction formulae themselves.

Main findings: (1) m_e and m_μ/m_e are the least model-dependent anchors, but m_μ/m_e is QED-dependent and derives from a 27-year-old muonium HFS measurement whose theory uncertainty CODATA underestimates by a factor of 5–10 (Eides 2026). (2) α carries an unresolved 5σ tension between the two best recoil measurements. (3) $v = 246$ GeV is never measured; it is the SM definition $v \equiv (\sqrt{2}\,G_F)^{-1/2}$. (4) G_F carries a theory correction $\Delta q = -0.44\,\%$ with a 4.7σ hadronic tension. (5) m_τ has two measurement methods differing by 0.18 MeV; FFGFT lies 0.4σ from the PDG average. (6) The magnitude of all theory uncertainties $(10^{-7}$

to 10^{-10}) does not explain the $\sim 1\,\%$ residual in $(m_\mu/m_e)^2$ — it remains physical. (7) Errors accumulate along the derivation chain; every prediction requires its full provenance path.

All 21 primary sources were verified on 5 September 2026.

1 The question

The corpus compares FFGFT predictions with PDG values and finds residuals at the sub-percent to percent level. Are PDG values theory-neutral, or do they themselves depend on the Standard Model — such that part of the residual arises on the extraction side rather than the FFGFT side?

The FFGFT foundational statement: only ratios free of irrational numbers are exact; SI absolute values are structurally approximate (Docs. 241, 085). If PDG values themselves contain irrational factors in their extraction, the same limitation applies to the comparison side.

2 How empirical values arise

No PDG or CODATA value is a raw measurement. Each passes through the same chain:

raw observable → theory formula → extracted parameter → CODATA adjustment → PDG value

The CODATA adjustment

CODATA publishes recommended values every four years, produced by a joint least-squares adjustment over roughly 300 input data linked by theory relations (QED, Rydberg formula, kinematics). The values are therefore *correlated*: a change in one input shifts all dependent values. Where theory uncertainties enter, they are partly treated as adjustment parameters (δ_{th} in the observational equations) rather than as errors — this is the Eides finding for muonium HFS **[Q]**.

Electron mass m_e

Raw observable: cyclotron frequency of a hydrogen-like ion ($^{12}C^{5+}$, $^{28}Si^{13+}$, $^{20}Ne^{9+}$) in a Penning trap, and the Larmor precession frequency of the bound electron. The ratio is a pure number.

Theory formula: bound g-factor from QED bound-state theory. From the frequency ratio and g_{bound} follows m_e/m_{ion}; with the ion mass in u follows m_e in u; conversion to MeV via u → kg → MeV.

Theory dependence: QED for g_{bound}, small and well controlled. Current: Heiße et al. (2023) at 0.1 ppb **[Q]**.

Muon-electron mass ratio m_μ/m_e

Raw observable: ground-state hyperfine splitting of muonium — a microwave resonance at 4.463 GHz — and the Zeeman splitting in a magnetic field.

Theory formula: $\Delta\nu_{\text{HFS}} = \nu_F\,[1 + \text{corrections}]$ with the Fermi frequency

$$\nu_F = \frac{16}{3}\,\alpha^2\,c\,R_\infty\,\frac{m_e}{m_\mu}\left(\frac{m_r}{m_e}\right)^3. \tag{1}$$

Corrections: QED to $O(\alpha^3)$, recoil in m_e/m_μ, weak Z exchange, hadronic vacuum polarisation.

Theory dependence: high. Sole precision measurement: LAMPF 1999 (Liu et al.). Eides (2026) shows that CODATA lists the measured value 4 463 302 776(51) Hz as the theoretical value; the actual theory uncertainty is 271–515 Hz **[Q]**.

For FFGFT: $\nu_F \propto \alpha^2$. The anchor m_μ/m_e implicitly contains α. Anchor and α are not independent.

Fine-structure constant α

Two routes that disagree at 5σ **[Q]**:

- **Atom recoil:** $\alpha^2 = 2R_\infty\,(h/m_{\text{atom}})\,(m_{\text{atom}}/m_e)/c$. Rb (Morel et al. 2020): $\alpha^{-1} = 137.035\,999\,206(11)$. Cs (Parker et al. 2018): $137.035\,999\,046(27)$. Difference 1.6×10^{-7}, $> 5\sigma$.
- **Electron $g-2$:** a_e at 10^{-13} (Fan et al. 2023) plus 5-loop QED (12 672 diagrams, Aoyama et al.). Gives $\alpha^{-1} = 137.035\,999\,166(15)$ — between Rb and Cs.

The CODATA value 137.035 999 177(21) is an average covering none of the measurements. FFGFT: $\alpha^{-1} = 3700/27 = 137.037\,037\ldots$ (Doc. 338, R101), deviation 7.6 ppm — 100× larger than the Rb/Cs tension.

Tau mass m_τ — two methods

Method	Experiment	Value (MeV)	Theory assumption
Threshold scan	BESIII 2014	$1776.91 \pm 0.12\,^{+0.10}_{-0.13}$	QED cross-section
Pseudomass	Belle II 2023	$1777.09 \pm 0.08 \pm 0.11$	$m_{\nu_\tau} = 0$
PDG average 2024	—	**1776.93 ± 0.09**	—

Threshold scan: $\sigma(e^+e^- \to \tau^+\tau^-)$ versus beam energy near $2m_\tau \approx 3.55$ GeV; m_τ as fit parameter of the QED threshold curve. **Pseudomass:** at 10.6 GeV, the kinematic edge of the invariant mass in $\tau \to 3\pi\nu$; assumes $m_{\nu_\tau} = 0$.

For FFGFT: $m_\tau = 1776.97$ MeV (pred) lies 0.04 MeV $= 0.4\sigma$ from the PDG average. The methods differ by 0.18 MeV. The falsification test is undecided; Belle II with > 1 ab^{-1} should reach $\sigma < 0.05$ MeV.

Fermi constant G_F and vacuum expectation value v

Raw observable for G_F: muon decay time distribution (MuLan, 10^{12} decays), $\tau_\mu = 2.196\,980\,3(22)$ µs **[Q]**.

Theory formula (Eberhart et al. 2026 **[Q]**):

$$\tau_\mu^{-1} = \frac{G_F^2\, m_\mu^5}{192\pi^3}\,(1+\Delta q), \qquad \Delta q = (-4\,384\,678 \pm 34)\times10^{-9} \approx -0.44\,\%. \tag{2}$$

Δq contains QED to $O(\alpha^3)$, hadronic vacuum polarisation, electron-mass effects. Purely electroweak corrections are absorbed into G_F by definition (Sirlin 1980). The hadronic component carries a 4.7σ tension (CMD-3 vs. older data).

Raw observable for v: none. $v \equiv (\sqrt{2}\,G_F)^{-1/2} = 246.22$ GeV follows from $M_W = gv/2$ and $G_F/\sqrt{2} = g^2/(8M_W^2)$. v **exists only within the Standard Model.** Its numerical value is the muon lifetime in SM clothing.

For FFGFT: Doc. 006/R104 gives $v = 248.3$ GeV (without K_{frak}) or 245.6 GeV (with $K_{\text{frak}} = 74/75$). Comparison with 246.22 GeV is a comparison of *FFGFT structure against SM definition*, not against an observable. Only τ_μ and m_μ are measured.

Gravitational constant G

Torsion balance (Li et al. 2018, 11.6 ppm) or atom interferometry (Rosi et al. 2014, 150 ppm). Theory: Newton. The only model-free value, but the least precise: CODATA $6.67430(15)\times10^{-11}$, 22 ppm **[Q]**.

Neutrino mass-squared differences

Event rates versus L/E, three-flavour fit (NuFIT 6.0 **[Q]**). Δm^2_{31} depends on whether SK atmospheric data are included: ratio 33.7 (with) vs. 33.9 (without). FFGFT $360/11 = 32.73$; measurement uncertainty 1–3 % ≈ FFGFT residual. Not discriminating.

Overview

Quantity	Raw observable	Theory in extraction	Convention
m_e	frequency ratio	QED (g_{bound})	—
m_μ/m_e	HFS frequency	QED, weak, hadr.	—
α (Rb/Cs)	interferometer phase	R_∞ (QED)	—
α ($g-2$)	frequency ratio	5-loop QED	—
m_τ	rates vs. E / edge	QED threshold / $m_{\nu_\tau}=0$	beam calibration
G_F	decay times	Fermi + QED + hadr.	Sirlin absorption
v	—	—	$v \equiv (\sqrt{2}G_F)^{-1/2}$
G	angle / acceleration	Newton	—
Δm^2	rates vs. L/E	3-flavour	with/without SK-atm

No value is theory-free. For FFGFT this means: comparison with PDG values is comparison with "QED + convention", not with raw observations.

3 Irrational factors in the extraction formulae

The FFGFT foundational statement — ratios of rationals are exact, SI absolute values carry the approximation of the projection $S^1_m \to \mathbb{R}_t$ (Docs. 306, 333) — applies to the SM extraction formulae themselves:

Quantity	Formula	Irrational factor
G_F	$\tau_\mu^{-1} = G_F^2 m_\mu^5/(192\pi^3)$	π^3
v	$v = (\sqrt{2}\,G_F)^{-1/2}$	$\sqrt{2}$
α (recoil)	$\alpha^2 = 2R_\infty h/(mc)$	$R_\infty \propto \alpha^2$: implicitly circular
HFS	$\nu_F \propto \alpha^2$	α^2
Δq	expansion in α/π	π in every term

"G_F measured" means: τ_μ measured, divided by $192\pi^3$, square root taken. "v measured" additionally means: divided by $\sqrt{2}$, square root again. The irrational factors come from the SM formulation (phase-space integral, Higgs normalisation), not from measurement. The only convention-free comparison would be at the level of $\tau_\mu = 2.196\,980\,3$ µs — which FFGFT would have to predict directly **[S]**.

4 The residual in $(m_\mu/m_e)^2$: exact computation

FFGFT bare (Doc. 338): $(m_\mu/m_e)^2 = 43200$ exactly, an integer, $= |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|$. With CODATA 2022 in exact rational arithmetic:

$$(m_\mu/m_e)^2_{\text{PDG}} = 42753.12 \tag{3}$$

$$\text{residual} = 446.88 \quad (1.045\,\%) \tag{4}$$

$$\sigma_{\text{rel}}\big((m_\mu/m_e)^2\big) = 4.35 \times 10^{-8} \quad \Rightarrow \quad \text{residual} = 240\,000\,\sigma. \tag{5}$$

Possible error sources, computed:

- CODATA uncertainty: ± 0.002 — factor 2×10^5 too small.
- Eides correction to HFS theory uncertainty (515 instead of 51 Hz): ± 0.010 — residual remains $45\,000\,\sigma$.
- Choice of α in the HFS anchor (FFGFT $3700/27$ instead of CODATA): shifts $(m_\mu/m_e)^2$ by -1.3 — factor 500 too small.
- To reach 43200, m_μ would need to be 551 keV higher (measurement error 2.3 eV), or m_e 2.65 keV lower (error 0.15 meV), or the HFS 23 MHz lower (error 51 Hz).

Conclusion [B]: No measurement error explains the residual. It is physical and belongs to the theory: fractal-recursive correction (Docs. 295, 146, 338) and projection approximation (Docs. 306, 333). In the square, without square root: $43200 \cdot (1 - 77.6\,\xi) = 42753$; $K_{\text{frak}} = 1 - 100\xi$ over-corrects by $22.4\,\xi$. Which power of K_{frak} acts on the quadratic identity is not fixed in the corpus **[S]**.

Rational or irrational: where the comparison is clean

The discussion above compared three Galois identities against PDG values. Not all comparisons are equally clean:

Comparison	Galois value	Type	Residual
$(m_\mu/m_e)^2$ vs. 43200	43200 (integer)	rational	$-1.034\,\%$
m_μ/m_e vs. $\sqrt{43200}$	$120\sqrt{3}$	**irrational!**	$-0.519\,\%$
$m_e \cdot m_\mu$ vs. 54 MeV^2	54 (integer)	rational	$-0.016\,\%$

The linear comparison m_μ/m_e vs. $\sqrt{43200} = 120\sqrt{3}$ is *not a clean rational comparison*: $\sqrt{43200}$ is irrational, and the $-0.52\,\%$ residual is not an independent physical finding. It equals exactly half the quadratic residual ($-1.034\,\%/2 \approx -0.517\,\%$), because taking the square root approximately halves the deviation. Nothing new becomes visible.

Only two comparisons are algebraically clean:

- **Square:** $(m_\mu/m_e)^2 = 43200$ — both sides rational, residual $-1.034\,\%$ physical.
- **Product:** $m_e \cdot m_\mu = 54$ MeV^2 — both sides rational, residual $-0.016\,\%$ **[K]** (Doc. 338; algebraic origin: $|GF(3)^*| \cdot |GF(27)| = 2 \cdot 27 = 54$).

The product $m_e \cdot m_\mu = 54$ MeV^2 is the algebraically cleanest Galois identity with the smallest rational residual ($-0.016\,\%$). The absolute deviation is -0.0087 MeV^2 ($-7400\,\sigma$) — also physical, but 65 times smaller than the quadratic residual.

Bookkeeping rule: Galois identities should be compared only in their natural algebraic form — where no irrational numbers appear. For $(m_\mu/m_e)^2 = 43200$ that is the square comparison; for $m_e \cdot m_\mu$ the product. The linear comparison m_μ/m_e vs. $\sqrt{43200}$

should be avoided because it introduces an irrational intermediate (verification script: `pruef_351_rationale_vergleiche.py`).

The residual of the product comparison lies at order ξ itself:

Quantity	Residual in units of ξ	Interpretation
$m_e \cdot m_\mu$ vs. 54	$-1.21 \cdot \xi$	order ξ: SI projection **[K]**
$(m_\mu/m_e)^2$ vs. 43200	$-77.6 \cdot \xi$	higher order: fractal-recursive **[K]**

The SI projection $S^1_m \to \mathbb{R}_t$ (Docs. 085, 306, 333) structurally produces errors of order ξ. The product residual $-0.016\,\% \approx 1.2 \cdot \xi$ lies precisely in this range and is therefore structurally explainable as an SI transition approximation **[K]**. The quadratic residual $-1.034\,\% \approx 78 \cdot \xi$ lies in higher order and is explained by the fractal-recursive correction (Docs. 295, 146, 338).

Result: The Galois identity $m_e \cdot m_\mu = 54$ MeV^2 is the only currently known identity whose residual lies at order ξ — the structurally expected SI projection error. The residual is therefore not unexplained; it corresponds to the approximation of the SI transition.

Constitutional description (after Stefaan Vossen, 5 Sept. 2026)

The following four-layer formulation summarises the epistemic status of the FFGFT comparisons:

Layer	Status
Layer-1	Exact within rational arithmetic: $(m_\mu/m_e)^2 = 43200$ and $m_e \cdot m_\mu = 54$ MeV^2 are integer Galois identities without approximation **[B]**.
Layer-2	One empirical dimensional anchor: $m_\mu = 105.66$ MeV (PDG). All absolute predictions follow from it **[K]**.
Residual	Empirical discrepancy with a structural explanation of approximately the correct order: $(m_\mu/m_e)^2$ deviates by $\approx 78 \cdot \xi$ from the PDG value (fractal-recursive correction, Doc. 295); $m_e \cdot m_\mu$ deviates by $\approx 1.2 \cdot \xi$ (SI projection approximation, Docs. 085, 306, 333) **[K]**.
Quantitative bridge	The forward derivation of the exact residual from the $S^1_m \to \mathbb{R}_t$ non-isomorphism is not in the corpus. **Open [S].**

The open [S] is not a gap in the pejorative sense but a precisely constituted limit: the theory states explicitly where its formal construction stops.

5 Error accumulation along the derivation chain

Each step carries a small residual (recursion $\sim 100\xi$, projection $\sim \xi$, extraction theory 10^{-7}–10^{-10}). Already at m_μ, the first step after the anchor m_e:

Route $m_e \to m_\mu$	Residual
Galois bare: $m_\mu = \sqrt{43200}\, m_e$	$+0.52\,\%$
with $\sqrt{K_{\text{frak}}}$: $m_\mu = \sqrt{43200 \cdot 74/75}\, m_e$	$-0.15\,\%$
via α: $E_0 = \sqrt{\alpha/\xi}$, $m_\mu = E_0^2/m_e$	$+1.37\,\%$

At m_τ (second step), v (third, via $\xi^{3/2}$), each derivation accumulates further. To decide where a residual sits, convergence of empirical values onto the algebraic ones would have to be established — it is not: m_μ/m_e (1999), α (5σ tension), Δm^2_{atm} (2.4σ shift in 2024), v (not measured).

Bookkeeping: For every reported prediction, state: (1) the anchor, (2) the number of derivation steps, (3) corrections applied and where, (4) the comparison value with its extraction method. "FFGFT predicts m_μ to 0.5 %" is incomplete; complete: "From m_e (PDG), in one step via the Galois identity, without K_{frak}, m_μ follows with $+0.52\,\%$ against the 1999 muonium HFS value."

6 Consequences for the corpus

No numerical value in the corpus changes. What changes:

1. **v is neither anchor nor observable.** Docs. 006, 319, R104: "$v = 246$ GeV (measured)" → "$v = 246$ GeV (SM definition from G_F)". Register R112.
2. **m_μ/m_e is QED-dependent.** "model-independent" → "least model-dependent available value"; $v_F \propto \alpha^2$. Register R113.
3. **α has no unique measured value below 1 ppb.** Cite the comparison value as CODATA average with Rb/Cs tension.
4. **m_τ:** two methods, PDG average 1776.93(9), FFGFT 0.4σ — open.
5. **New open item [S]:** τ_μ directly from FFGFT, without detour via G_F and v — the only convention-free test of the weak sector.

Sources

All sources verified on 5 September 2026 against INSPIRE, APS, Nature, arXiv, or PubMed. Numerical values computed in exact rational arithmetic against CODATA 2022.

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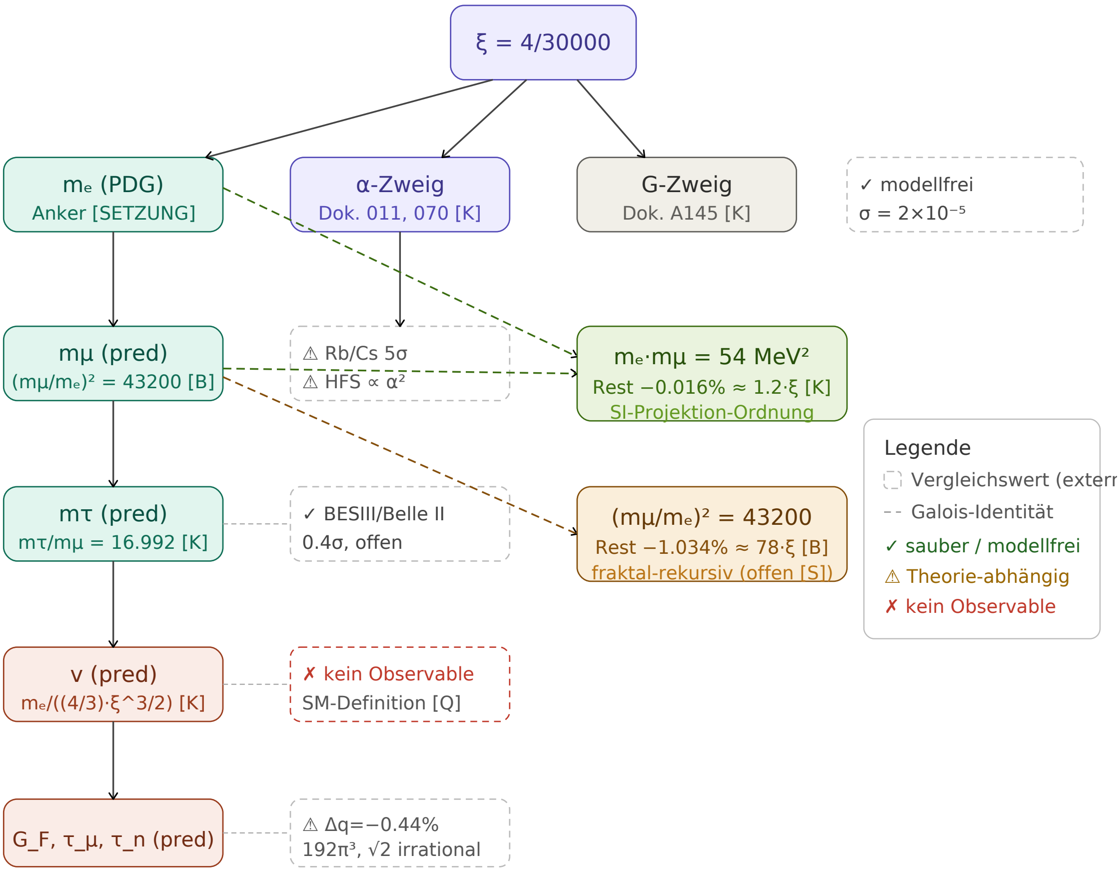

Derivation chains: from ξ to comparison values. Dashed boxes = external comparison values; dashed arrows = Galois identities.

1 Affected derivation chains

Which FFGFT chain depends on which comparison value, and what each finding changes.

Finding	Affected documents	Numerical correction	Wording correction
v is SM definition (R112)	Docs. 006, 319, R104	none	"$v = 246$ GeV (measured)" → "(SM definition from G_F)"
G_F: Δq, $192\pi^3$, $\sqrt{2}$	Docs. 006, R104, 319	none	state $\Delta q = -0.44\,\%$; clarify with/without Δq
m_μ/m_e QED-dependent (R113)	Docs. 338, 317, 011	none	"model-independent" → "least model-dependent"
α Rb/Cs 5σ	Docs. 011, 070, 338	none	cite CODATA average with tension
m_τ two methods	Docs. 317, 006	none	cite BESIII + Belle II; PDG average 1776.93(9)
Neutrinos Δm^2	Docs. 339, 340	none	freeze NuFIT configuration
G	Doc. A145	none	—
Total		**none**	wording clarifications

The residual $(m_\mu/m_e)^2 = 43200$ vs. 42753.12 (difference 446.9) is $240\,000\,\sigma$ of the measurement uncertainty — physical (R113). No numerical value in the corpus changes.